LESSON PLAN

PYRO TECHNIQUES (34)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Power supply system –purpose and requirements
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Power supply system –purpose and requirements
5. Specific Objectives	34.1.1.1 Trainee will be able to explain various terms on guided weapon.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of power supply block diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a power supply system in a guided weapon?
(b) List three essential requirements for a weapon power supply.
(c) Define the term 'voltage regulation' in the context of guided weapons.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of power supply system • Voltage and current requirements • Stability and reliability needs • Integration with guidance electronics	L & Ds	PPT, IB	15 min
(b)	• Battery types used in missiles • Power conditioning units • Distribution networks • Safety isolation mechanisms	L & Ds	PPT, PDF	20 min
(c)	• Failure modes of power systems • Redundancy requirements • Maintenance access points • Testing procedures for power units	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how power supply requirements differ between the launch phase and flight phase.
(b) Describe the role of redundancy in ensuring power supply continuity for guidance systems.

12. Home Assignment and Follow Up	Prepare a brief note on the specific voltage and current requirements for the Dhanush missile power supply system to be discussed in LP 09.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction, purpose & leading particulars of Dhanush missile.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min